situation, Zel'dovich asserted, the rapidly spinning body will grab hold of the fluctuational wave and accelerate it, much like a small boy accelerating a slingshot as he swings it faster and faster. The acceleration feeds some of the body's spin energy into the wave, amplifying it. The new, amplified portion of the wave is a "real wave" with positive total energy, while the original, unamplified portion remains a vacuum fluctuation with zero total energy (Box 12.4). The spinning body has thus used the vacuum fluctuation as a sort of catalyst for creating a real wave, and as a template for the shape of the real wave. This is similar, Zel'dovich pointed out, to the manner in which vacuum fluctuations cause a vibrating molecule to "spontaneously" emit light (Box 12.4).

Zel'dovich told me he had proved that a spinning metal sphere radiates in this way; his proof was based on the laws of *quantum electrodynamics*—that is, the well-known laws that arise from a marriage of quantum mechanics with Maxwell's laws of electromagnetism. Though he did not have a similar proof that a spinning black hole will radiate, he was quite sure by analogy that it must. In fact, he asserted, a spinning hole will radiate not only gravitational waves, but also electromagnetic waves (photons[9]), neutrinos, and all other forms of radiation that can exist in nature.

I was quite sure that Zel'dovich was wrong. Several hours later, with no agreement in sight, Zel'dovich offered me a wager. In the novels of Ernest Hemingway, Zel'dovich had read of White Horse scotch, an elegant and esoteric brand of whisky. If detailed calculations with the laws of physics showed that a spinning black hole radiates, then I was to bring Zel'dovich a bottle of White Horse scotch from America. If the calculations showed that there is no such radiation, Zel'dovich would give me a bottle of fine Georgian cognac.

I accepted the wager, but I knew it would not be settled quickly. To settle it would require understanding the marriage of general relativity and quantum mechanics far more deeply than anyone did in 1971.

Having made the wager, I soon forgot it. I have a lousy memory, and my own research was concentrated elsewhere. Zel'dovich, however, did not forget; several weeks after arguing with me, he wrote down his argument and submitted it for publication. The referee probably would have rejected his manuscript had it come from somebody else; his argument was too heuristic for acceptance. But Zel'dovich's reputation carried the day; his